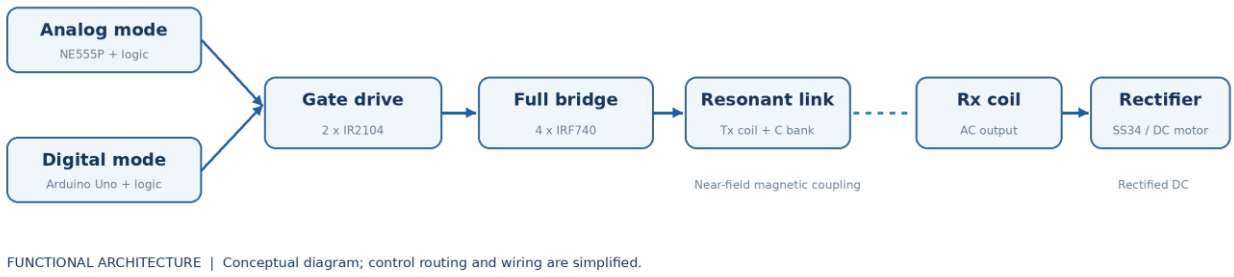
ENGINEERING PROJECT

DESIGN AND CIRCUIT ANALYSIS OF A RESONANT WIRELESS POWER TRANSFER SYSTEM

Dual-mode NE555P / Arduino Uno excitation | IR2104 full-bridge drive | DC motor load



Functional overview of the proposed wireless power transfer system.

Abstract

This report documents the topology and intended operation of a short-range resonant wireless power transfer prototype. A low-voltage control section generates switching commands in either analog mode, using an NE555P-based oscillator, or digital mode, using an Arduino Uno. Two IR2104 high-side/low-side drivers operate four IRF740 MOSFETs arranged as a full bridge. The bridge excites a transmitter coil through a selectable capacitor network. Energy is magnetically coupled to a separate receiver coil, rectified by an SS34 Schottky-diode bridge, filtered, and delivered to a DC motor as the demonstration load.

The analysis is based on five supplied schematic views and the project functional description. It provides an integrated block-level explanation, traces the relevant circuit stages, introduces the governing resonance and power-transfer relationships, and identifies schematic-level verification items. Because measured coil parameters, oscillograms, electrical power data, firmware, and the demonstration video were not supplied for review, no numerical transfer efficiency, switching frequency, coil separation, or experimental power rating is asserted. The reported observation that the motor operates wirelessly is presented as project-reported functionality rather than independently verified test data.

Keywords

Wireless power transfer; inductive coupling; resonant coupling; full-bridge inverter; H-bridge; IR2104; IRF740; NE555P; Arduino Uno; DC motor; Schottky rectifier.

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> **Scope of evidence:** Component values are transcribed only where legible in the schematic views. Functional descriptions are reported project intentions; numerical performance remains unverified unless explicitly stated.

1. Introduction and Design Objectives

Wireless energy delivery over a short air gap can be achieved through a time-varying magnetic field linking two electrically isolated coils. Unlike a wired connection, the energy path does not require metallic contact between the transmitting and receiving assemblies. The central engineering challenge is to produce a sufficiently strong alternating magnetic field, tune the electrical network for the selected operating frequency, and efficiently recover usable DC power at the receiver.

In this project a full-bridge MOSFET inverter converts a DC supply into a bidirectional excitation for the transmitting coil. The alternating field induces voltage in a second coil. A diode bridge rectifies this voltage, and a smoothing capacitor reduces the ripple presented to a small DC motor. The project description reports that the motor runs when the transmitter and receiver are appropriately coupled.

1.1 Intended Design Objectives

- Implement a discrete four-MOSFET H-bridge to generate alternating excitation from a DC bus.

- Provide two alternative command sources: an adjustable NE555P analog oscillator and Arduino Uno digital control.
- Use high-side/low-side gate drivers with bootstrap circuitry for the MOSFET bridge.
- Tune the transmitter coil using selectable capacitors and measured coil inductance.
- Rectify received AC and operate a DC motor without a galvanic connection between coils.
- Document circuit behavior and the limits of conclusions supportable by the available design evidence.

1.2 What the Demonstration Establishes

Operation of a motor connected to the receiver, if observed under separated-coil conditions, demonstrates that electrical energy reached the load through inductive coupling. It does not by itself establish transfer efficiency, regulated output voltage, safety certification, long-range transfer capability, or a particular resonant quality factor. These require additional measurements.

2. System Architecture and Modes of Operation

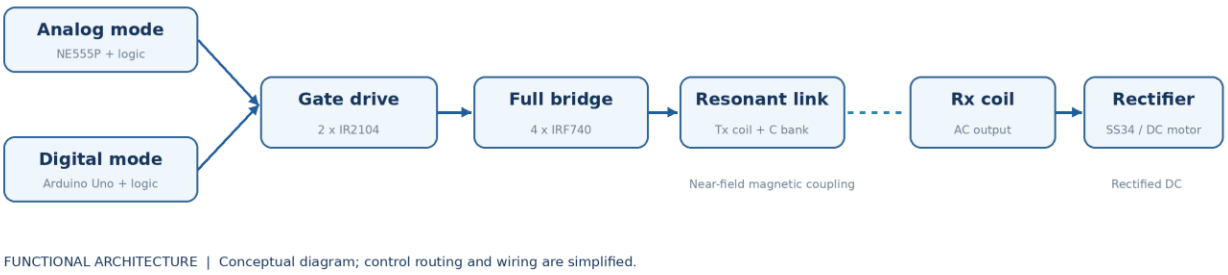


Figure 1. System-level functional diagram (explanatory; not an exact electrical netlist).

2.1 Energy Path

DC input -> IRF740 full-bridge inverter -> transmitter matching capacitor network and coil -> magnetic coupling through air -> receiver coil -> SS34 full-wave bridge -> DC smoothing capacitor -> motor load. The switching and power conversion stages are physically separated from the control logic through the driver architecture; nevertheless, the isolation of each power supply and coil interface must be verified against the complete assembled hardware.

2.2 Analog Control Mode

The NE555P timing circuit is configured as an oscillator. A 100-kohm potentiometer (R18) and two SS34 steering diodes (D3 and D4) alter the charging and discharging paths of its timing capacitor. This arrangement enables adjustment of pulse timing and, at a suitable setting, an approximately balanced duty cycle. A 7404 inverter and 7408 AND logic provide complementary / gated command paths prior to the IR2104 inputs. Exact operating frequency, duty ratio, and selection logic are not established from the screenshots alone.

2.3 Digital Control Mode

In digital mode, an Arduino Uno is intended to replace the NE555P as the switching-command source. The schematic shows digital input labels for two command channels (IN_1_D and IN_2_D), but the supplied material does not include the Arduino program, the complete selection wiring, or measured logic waveforms. The firmware must generate coordinated bridge-leg commands that maintain the intended polarity alternation and do not disable bootstrap refresh.

2.4 Why a Full Bridge Is Used

The four-switch topology provides both polarities of voltage across the transmitter connection. Under ideal bipolar switching, one diagonal pair produces approximately +Vdc across the load and the opposite diagonal produces approximately -Vdc. This alternating excitation establishes the changing magnetic flux needed for induction. The gate-driver outputs within each half bridge are complementary, but the two bridge legs must still be commanded with the proper relative phase for useful differential voltage.

3. Detailed Circuit Analysis

3.1 Receiver Coil, Full-Wave Rectifier, and DC Load

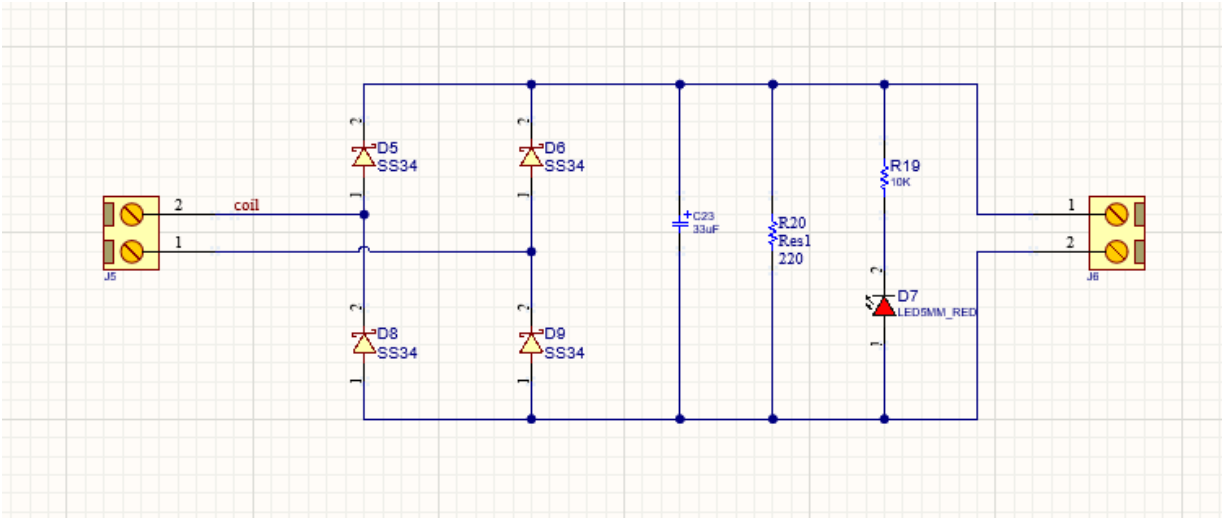


Figure 2. Receiver schematic: J5 coil interface; SS34 bridge D5, D6, D8, D9; C23; output J6; R19/D7 and R20.

Two AC terminals at J5 connect to a four-diode bridge formed by D5, D6, D8, and D9 (SS34). Its DC rails supply output connector J6. Capacitor C23 is labeled 33 uF and is connected across the DC rails, providing output smoothing. R19 (10 kohm) and red LED D7 form an indicator branch. R20 is labeled 220 ohm and is placed directly across the DC output rails, where it acts as a permanent parallel load.

> **Engineering observation:** R20 draws Vout/220 A and dissipates Vout^2/220 W if the schematic value is correct. For illustration, at 12 V this is about 55 mA and 0.65 W, independent of motor consumption. Its purpose and power rating need confirmation. The actual motor is connected through J6, not depicted as a distinct motor symbol.

3.2 Auxiliary Power Supply and Status Indicators

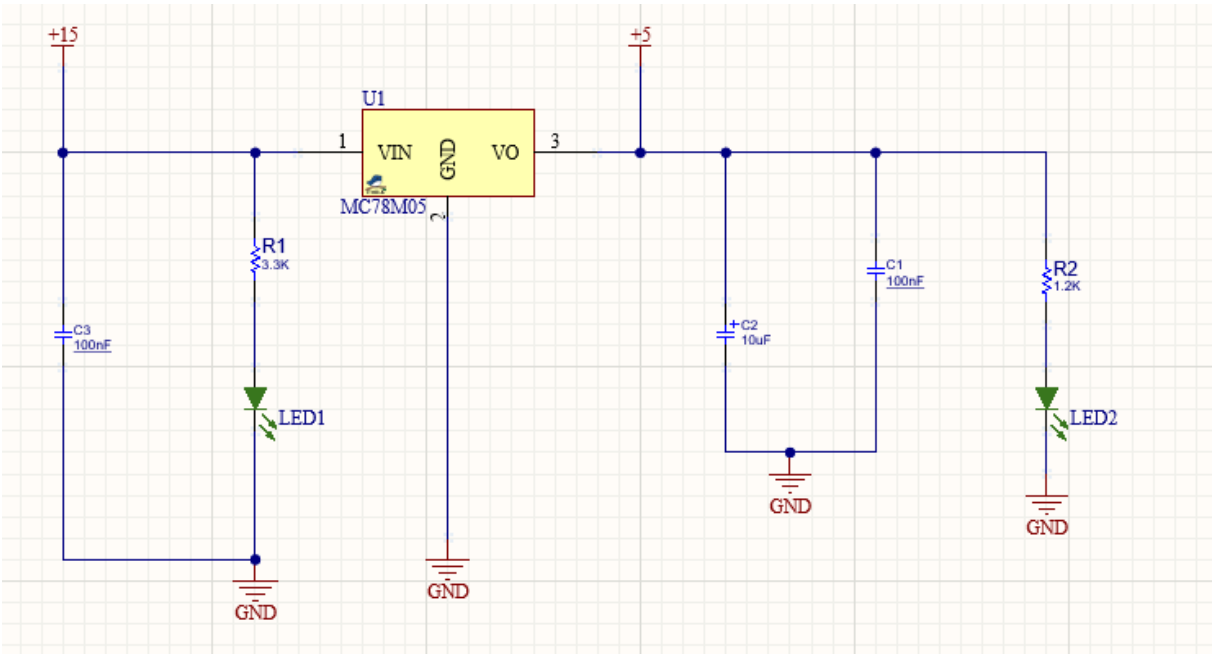


Figure 3. MC78M05 linear regulation from +15 V to +5 V, with decoupling and indicator LEDs.

The MC78M05 linear regulator U1 converts the indicated +15 V supply to a +5 V logic rail. C3 (100 nF) appears at the input; C2 (10 uF) and C1 (100 nF) provide output-side filtering. Separate LED branches indicate the presence of the supply rails. For an idealized linear regulator, the regulator dissipation is approximately (Vin - Vout) x Iload; at a 15 V to 5 V conversion, thermal feasibility depends strongly on the control-side current. Exact LED polarity, component power ratings, and supply-return routing should be checked against the native schematic/netlist.

3.3 MOSFET Full Bridge and IR2104 Gate Drivers

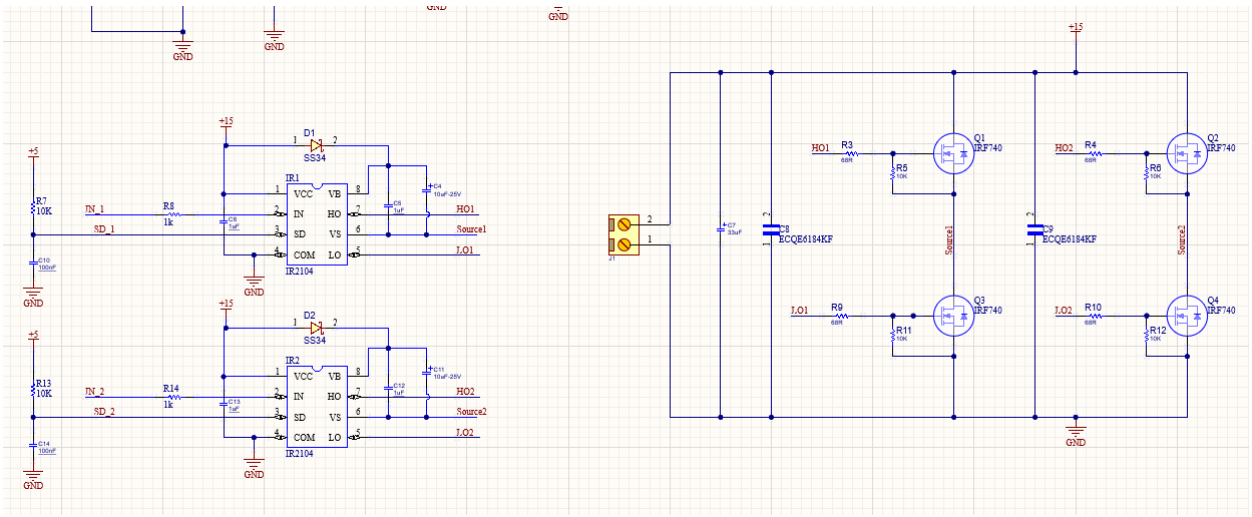


Figure 4. IR2104 driver stages and IRF740 bridge: IR1/IR2, D1/D2 bootstrap diodes, Q1-Q4 and source nodes.

Q1-Q4 are identified as IRF740 N-channel MOSFETs. Q1/Q3 form one half bridge and Q2/Q4 form the other; the midpoint nets are labeled Source1 and Source2. Each half bridge is driven by an IR2104, with HO (high side) and LO (low side) outputs. D1 and D2 are SS34 bootstrap diodes. Capacitors between VB and VS provide the floating high-side gate supply; the schematic also shows gate-series resistors labeled 68 ohm and 10-kohm gate-to-source discharge resistors. A +15 V bus appears at the positive rail, with ground at the negative rail.

The IR2104 provides complementary high-/low-side control with an internally set dead time and an active-low shutdown input. These features reduce the likelihood of simultaneous high-/low-side conduction within one leg; correct bridge operation still requires sensible phase relationships, adequate bootstrap supply, and verification of startup/shutdown sequencing. Manufacturer data are cited in the references.

> **Device-selection concern:** The IRF740 is a relatively high-voltage MOSFET with a published maximum RDS(on) of 0.55 ohm at VGS = 10 V. It can be functionally suitable for demonstration, but its conduction loss deserves attention in a low-voltage, current-bearing bridge. This is a datasheet-based design-review observation, not a measured loss value.

3.4 NE555P Oscillator and Control Logic

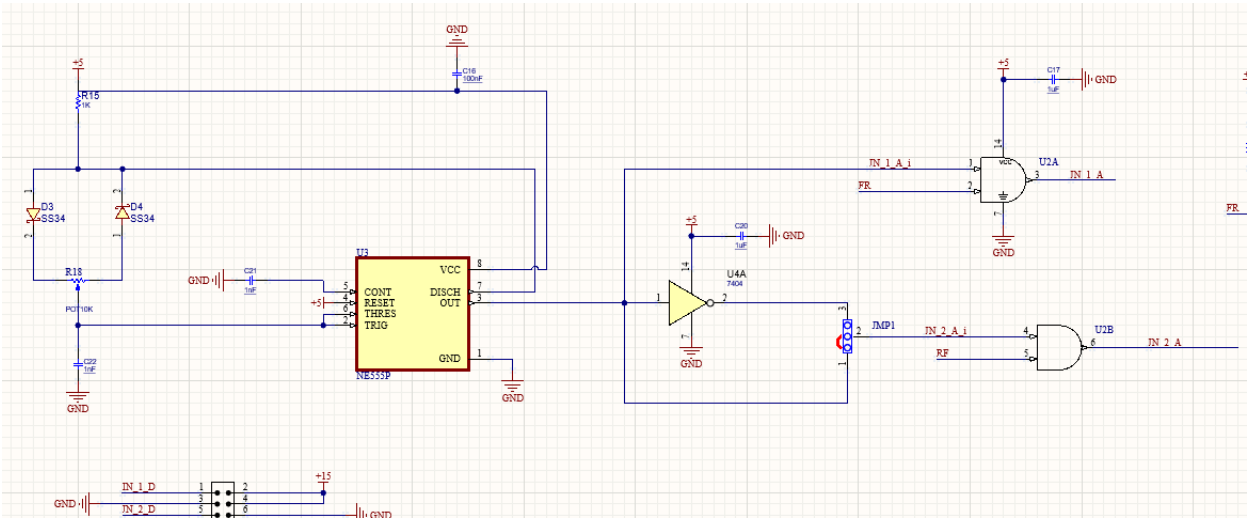


Figure 5. NE555P timing circuit, D3/D4 timing-path diodes, R18 potentiometer, 7404 inverter and 7408 control logic.

U3 is an NE555P timer supplied from +5 V. The timing network contains R15 (1 kohm), potentiometer R18 (100 kohm), diodes D3 and D4 (SS34), and timing capacitor C22 (1 nF as labeled). The output passes into a 7404 inverter U4A and two AND-gate branches U2A/U2B (7408). The logic nets RF, IN_1_A and IN_2_A and a jumper JP1 are visible. This arrangement is consistent with analog signal generation, phase inversion, and logical gating, but the screenshot does not establish a final truth table for all switch positions.

The project description describes an oscillator duty cycle of 50%. In practice the duty ratio depends on the potentiometer position and the component tolerances; the screenshot alone cannot confirm an exact 50% value. Similarly, adjusting the potentiometer changes the oscillator timing and therefore can change resonance alignment and delivered power; it should not be conflated with a directly measured, fixed motor-speed law.

3.5 Primary-Coil Compensation and Command Selection

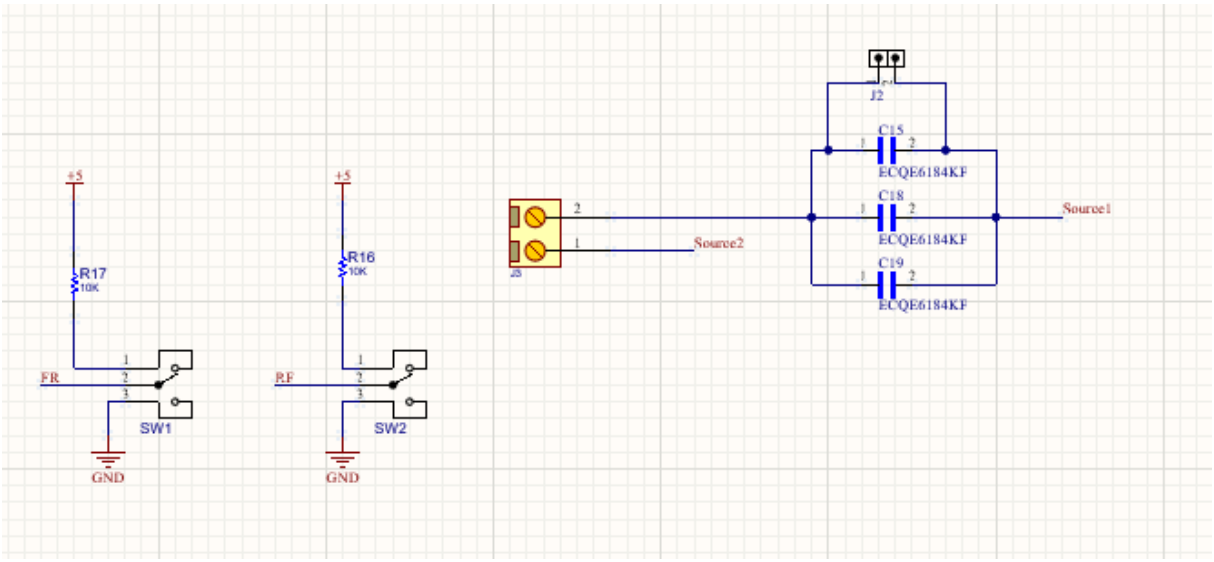


Figure 6. Source1/Source2 coil interface, selectable capacitor bank C15/C18/C19, bypass jumper J2, and mode/control switches.

The transmitter connector J3 and the Source1/Source2 nets define the primary-side coupling interface. Three capacitors labeled C15, C18, and C19 are arranged in parallel between a coil-terminal net and Source1, forming a selectable aggregate series-compensation element when installed as depicted. Jumper J2 bridges that network and appears capable of bypassing the series capacitance. Therefore the effect of J2 should be confirmed before any resonance experiment. SW1 and SW2 include +5 V pullups and connect a control signal to ground when actuated; their exact role is best verified using the native design connectivity.

The part-number text on the film capacitor symbols is visible, but an authoritative capacitance and tolerance cannot be inferred with confidence from the supplied screenshots alone. The true capacitance should be taken from the component BOM or physical capacitor markings and verified with measurement.

4. Resonant Coupling and Engineering Calculations

4.1 Fundamental Resonance Condition

A first-order series-resonant approximation is f0 = 1 / (2 pi sqrt(L C)), where L is the effective transmitter-coil inductance and C is the intended series-compensation capacitance. Rearranged for design, C = 1 / ((2 pi f0)^2 L). This provides an initial capacitor target after measuring L and choosing f0. The actual coupled system is affected by mutual inductance, coil resistance, source impedance, parasitic capacitance, and the nonlinear rectifier/motor load.

When C15, C18, and C19 are truly connected in parallel and the bypass is open, their equivalent capacitance is C_eq = C15 + C18 + C19. Closing a bypass jumper across this bank is not simply a small adjustment of f0; it can effectively remove that compensation impedance from the transmission path. Coil connection orientation and jumper behavior should therefore be verified from the native schematic before declaring the resonance mode.

4.2 Inductive Link and Coupling Coefficient

For coupled coils, the mutual inductance M can be written as M = k sqrt(Lp Ls), where Lp and Ls are primary and secondary inductances and k is the coupling coefficient (0 <= k <= 1 for the magnitude under usual definitions). The induced secondary voltage scales with angular frequency and linked flux, and may be approximated by Vs = j omega M Ip for sinusoidal steady-state analysis using a consistent polarity convention. Coil spacing, alignment, size, and frequency affect the useful magnetic coupling.

4.3 Voltage Conversion and Load Dependence

A full-wave bridge rectifies both polarities of the receiver AC voltage. The DC value depends on the received waveform, two diode drops in the conducting path, the smoothing capacitor, and the motor-current waveform. The motor is neither a pure resistance nor a constant-power load: startup current and back-electromotive force can materially change receiver voltage and link loading. Numerical output voltage cannot be computed reliably from the present evidence.

4.4 Power and Efficiency Definitions

An experimentally defensible efficiency figure requires simultaneous input and output power measurements: eta_link = P_out / P_in x 100%. For DC input, P_in can be approximated from mean DC-bus voltage and mean current only when ripple and measurement method are properly considered. On the output, P_out should include the motor power and, for a full receiver-power balance, any auxiliary resistive branches such as R20 and the indicator LED. The distinction between end-to-end motor power and total receiver output power should be kept explicit.

5. Demonstration and Verification Plan

5.1 Reported Operating Result

The project description reports an operating demonstration in which the transmitter bridge excites one coil, a nearby receiver coil picks up magnetic energy, and the rectified supply drives a DC motor. A demonstration video is reported to exist; it was not included in the files reviewed for this report. Consequently, no visual observation, measured spacing, rpm, waveform amplitude, output voltage, or efficiency is attributed to that video.

5.2 Suggested Verification Matrix

Test	Quantity / method	Evidence status
5-V rail	Measure U1 output and ripple under load	Not supplied
Oscillator	Measure NE555P frequency, duty and adjustment range	Not supplied
Bridge drive	Capture IN, HO, LO, Source1 and Source2 waveforms	Not supplied
Resonance	Measure L, effective C, frequency, and coil current	Not supplied
Receiver	Measure AC coil voltage and J6 DC voltage/current	Not supplied
Motor	Record starting behavior, operating current and coil gap	Reported qualitatively
Efficiency	Simultaneous DC source and receiver/motor power measurement	Not supplied

5.3 Suggested Measurement Sequence

- Verify +5 V and +15 V rail regulation, polarity, and no-load currents before connecting the power stage.
- Observe the digital or oscillator commands at logic level and confirm the expected driver-disable behavior.
- Check the high-side/low-side gate timing using appropriate high-voltage differential measurement techniques and correct probe grounding.

- Measure both bridge midpoints and their differential voltage at limited input current; observe any abnormal heating.
- Measure coil inductances and verify the installed matching-capacitor values and jumper positions.
- Gradually evaluate the coupled link with appropriately current-limited equipment, documenting output voltage and motor response.
- Log repeated tests at defined coil spacing and alignment; calculate efficiency only from corresponding measured power values.

6. Design Review and Engineering Considerations

The following items are design-review findings or verification requirements, not claims of fabrication error. They are prioritized because the supplied views alone do not disclose all electrical and physical design details.

Priority	Observation	Recommended verification
High	IRF740 conduction losses may be significant at low bus voltage.	Measure bridge current and device temperature; compare to device loss estimates.
High	Gate-driver bootstrap requires a valid refresh interval.	Capture VB-VS and HO-VS waveforms for the actual switching pattern.
High	Driver gate timing across both half bridges is not documented.	Verify complementary commands, dead time and startup/shutdown behavior.
High	J2 appears to bypass the primary series-capacitor network.	Confirm connectivity and jumper setting before resonant operation.
Medium	R20 = 220 ohm shunts the receiver output.	Confirm intended load, wattage and effect on transferred power.
Medium	The NE555 analog duty ratio may depart from 50%.	Measure frequency and duty across potentiometer settings.
Medium	SS34 diode current, reverse voltage and heating depend on load.	Validate rectifier margins including motor startup transient.
Medium	No evidence of snubbers, current sensing or overcurrent protection is visible in the cropped views.	Check full native design and physical prototype before declaring protection absent.
Low	Source net naming and analog/digital selection wiring are partly cropped.	Export full Altium schematic/netlist and annotate mode-selection truth table.

6.1 Loss, Heating and EMC

Major loss mechanisms include MOSFET conduction and switching loss, IR2104 driver power, transmitter copper and core-equivalent losses (the coil construction is not specified), capacitor ESR, diode conduction, and mechanical/electrical losses of the motor. Bridge dv/dt and resonant coil current can also produce electromagnetic interference, particularly with long loop areas. Placement of DC-bus decoupling capacitors, power-return geometry, thermal management, and loop layout merit review in the native PCB data.

6.2 Limitations and Reporting Discipline

The supplied evidence consists of schematic screenshots and narrative descriptions. No native Altium files, measured waveforms, coil inductance values, exact resonance frequency, quantified transmission distance, bill of materials with tolerances, Arduino source code, or demonstration video were reviewed. This report therefore distinguishes the proposed architecture from experimentally confirmed performance. It is not a certification, compliance assessment, or final production-release review.

7. Conclusions

The circuit implements the central building blocks of a short-range inductively coupled power-transfer demonstrator: adjustable or microcontroller-generated switching commands; complementary IR2104 gate drive; a four-device IRF740 full-bridge inverter; a compensated primary coil connection; and a rectifier/filter stage feeding a DC motor. The topology creates a coherent path from DC electrical input to an alternating magnetic field and back to DC at the receiver.

A principal contribution of the design is the combination of two drive modes with practical, discrete power electronics. The central technical requirement is the coordinated relationship among switching frequency, resonant capacitance, coil inductance, magnetic coupling, and load demand. The circuit review identifies achievable next verification steps - particularly gate timing, actual capacitor values, device temperatures, rectifier loading, and quantitative power measurements - without overstating the evidence presently available.

Appendix A. Identified Components and Evidence

Reference / block	Visible component or role
U1	MC78M05 regulator, +15 V input to +5 V
U3	NE555P timer
R18	100-kohm potentiometer
U4A; U2A / U2B	7404 inverter; 7408 AND logic
IR1 / IR2	IR2104 half-bridge gate drivers
Q1-Q4	IRF740 MOSFETs
D1 / D2	SS34 bootstrap-path diodes
C15, C18, C19; J2	Primary matching capacitor bank and bypass jumper
D5,D6,D8,D9	SS34 full-wave receiver rectifier
C23; R20	33-uF receiver capacitor; 220-ohm output shunt
J5 / J6	Receiver coil input / DC output
Arduino Uno	Alternative digital command generator

References

[1] Infineon Technologies, "IR2104(S) - Half-Bridge Driver" datasheet. https://www.infineon.com/assets/row/public/documents/24/49/infineon-ir2104-ds-en.pdf

[2] Texas Instruments, "NE555 - Single Precision Timer", product information and datasheet. https://www.ti.com/product/NE555

[3] Vishay Siliconix, "IRF740 Power MOSFET" datasheet. https://www.vishay.com/docs/91054/91054.pdf

[4] onsemi / Fairchild, "MC78M05 0.5 A Positive Voltage Regulator" datasheet. https://www.onsemi.com/download/data-sheet/pdf/mc78m05-d.pdf

[5] Arduino, "UNO R3" hardware documentation. https://docs.arduino.cc/hardware/uno-rev3/

[6] Project source material: five schematic screenshots and the functional description supplied with this project.